Functional Annotation of PP_4348 (Q88EV4): Cystathionine β-Lyase of *Pseudomonas putida* KT2440

Summary

PP_4348 (UniProt Q88EV4) of *Pseudomonas putida* KT2440 encodes a cytoplasmic, pyridoxal-5′-phosphate (PLP)-dependent cystathionine β-lyase (MetC; EC 4.4.1.8). Its primary, physiological function is to catalyze the α,β-elimination reaction that cleaves the Cβ–S bond of L,L-cystathionine, yielding **L-homocysteine, pyruvate, and ammonia** (Rhea RHEA:13965). This is the penultimate step of *de novo* methionine biosynthesis via the transsulfuration branch, mapped by KEGG to the methionine biosynthesis module M00017 (aspartate ⇒ homoserine ⇒ methionine). The gene symbol PP_4348, the organism *P. putida* KT2440, the trans-sulfuration enzyme family assignment, and the diagnostic InterPro/Pfam domains (IPR006233 bacterial cystathionine β-lyase; IPR000277 Cys/Met-metabolism PLP-dependent enzyme; PF01053) all coincide with the biochemistry described below. The gene identity is unambiguous — the well-known *P. putida* methionine γ-lyase is a *distinct* gene (*mdeA*/P13254), not this locus.

Structurally, PP_4348 belongs to the fold-type I family of PLP-dependent enzymes (the "γ-family" of the Cys/Met-metabolism group). Like its extensively characterized *Escherichia coli* ortholog MetC, it is expected to assemble as a homotetramer of catalytically active dimers, with each ~364-residue subunit organized into three functional domains. The PLP cofactor is anchored as an internal aldimine (Schiff base) to an active-site lysine — **Lys186 in PP_4348**, corresponding to the experimentally validated catalytic Lys210 of *E. coli* MetC. A pairwise global alignment performed in this investigation shows PP_4348 is most similar to *E. coli* cystathionine β-lyase (43.9% identity) — higher than to cystathionine γ-synthase MetB (41.8%), MalY (41.3%), or *P. putida* methionine γ-lyase (41.1%) — and that every key active-site residue defined by the *E. coli* crystal structure is conserved one-to-one. This colinear, intact catalytic constellation demonstrates that PP_4348 is a functional, canonical cystathionine β-lyase rather than a degenerate homolog.

Physiologically, direct experimental work in *P. putida* (Vermeij & Kertesz, 1999) showed that the transsulfuration enzymes cystathionine γ-synthase and cystathionine β-lyase are expressed at low basal levels but are strongly upregulated during growth on cysteine as the sole sulfur source, while the organism's dominant methionine route is direct sulfhydrylation of O-succinylhomoserine. PP_4348 therefore provides the forward (cysteine → methionine) transsulfuration route producing homocysteine; because *P. putida* lacks cystathionine γ-lyase (no reverse transsulfuration), the enzyme acts only in the biosynthetic direction. Like other members of its family, MetC is catalytically promiscuous, also exhibiting cysteine-S-conjugate β-lyase activity (EC 4.4.1.13) and, in some bacterial homologs, alliinase and D-amino-acid–metabolizing activities; however, L-cystathionine β-elimination is the enzyme's physiological reaction. The enzyme operates in the cytoplasm, consistent with the absence of a signal peptide in Q88EV4 and the cytosolic nature of the transsulfuration pathway.

Gene / Protein Identity Verification

Attribute	Value	Source
UniProt accession	Q88EV4	UniProt
Locus / gene	PP_4348	UniProt / EMBL AAN69927.1
Organism	*Pseudomonas putida* KT2440 (PSEPK)	UniProt
Length	364 aa	UniProt
EC / reaction	4.4.1.8 · RHEA:13965	UniProt catalytic activity
Cofactor	PLP, Schiff base at **Lys186**	UniProt features
Family	Trans-sulfuration enzymes	UniProt similarity
Signatures	IPR006233 (**Cys_b_lyase_bac**, bacterial-CBL specific), IPR000277, PF01053, COG0626, SSF53383	InterPro/UniProt

Verification result: The gene symbol, organism, reaction, cofactor, and domain architecture are internally consistent and match authoritative family models. The bacterial-CBL–specific InterPro signature **IPR006233** and the annotated catalytic reaction directly support the "cystathionine β-lyase" assignment. **No ambiguity detected.** No source contradicts the assignment.

Key Findings

Finding 1 — PP_4348 is a PLP-dependent cystathionine β-lyase (MetC, EC 4.4.1.8)

The core functional assignment is that PP_4348/Q88EV4 catalyzes:

> **L,L-cystathionine** + **H_2O** → **L-homocysteine** + **pyruvate** + **NH_4^+** (Rhea RHEA:13965)

This reaction is annotated in UniProt with a PLP cofactor bound through a Schiff base to Lys186 (N6-(pyridoxal phosphate)lysine). The reaction is an α,β-elimination that cleaves the Cβ–S bond of cystathionine. InterPro assigns PP_4348 the bacterial-cystathionine-β-lyase-specific signature **IPR006233 (Cys_b_lyase_bac)** in addition to the broader **IPR000277 (Cys/Met-metabolism PLP-dependent enzyme)**, Pfam **PF01053 (Cys_Met_Meta_PP)**, eggNOG **COG0626**, and SUPFAM **SSF53383 (PLP-dependent transferases, fold-type I)**. The bacterial-CBL-specific signature is particularly diagnostic — it distinguishes cystathionine β-lyases from the closely related γ-synthases and γ-lyases of the same fold.

A pairwise global alignment performed in this investigation quantifies the family placement. PP_4348 (364 aa) is most similar to *E. coli* cystathionine β-lyase MetC (P06721) at **43.9%** aligned identity, exceeding its similarity to cystathionine γ-synthase MetB (41.8%), MalY (41.3%), and *P. putida* methionine γ-lyase (41.1%). Although these figures are close — reflecting deep conservation of the fold-type I scaffold across the Cys/Met family — the highest identity is to the β-lyase, and the diagnostic InterPro signature resolves any residual ambiguity.

The reaction chemistry is supported by literature. A 2024/2025 structural study of *Bacillus cereus* cystathionine β-lyase states that these enzymes "catalyze the conversion of cystathionine to homocysteine and pyruvate" PMID: 39644606, matching the UniProt reaction for PP_4348. Foundational structural work on *E. coli* CBL establishes that "Cystathionine beta-lyase (CBL) is a member of the gamma-family of PLP-dependent enzymes, that cleaves C beta-S bonds of a broad variety of substrates" PMID: 8831789 — consistent with PP_4348's PLP cofactor and β-elimination mechanism.

Finding 2 — PP_4348 operates in the transsulfuration branch of methionine biosynthesis and is induced by cysteine as sole sulfur source

The most decisive organism-specific evidence comes from Vermeij & Kertesz (1999), who directly measured transsulfuration enzyme activities in *Pseudomonas putida* S-313 PMID: 10482527. They reported that "The enzymes of the transsulfuration pathway (cystathionine gamma-synthase and cystathionine beta-lyase) were expressed at low levels in both pseudomonads but were strongly upregulated during growth with cysteine as the sole sulfur source." This establishes both the *presence* and the *sulfur-source–dependent regulation* of cystathionine β-lyase in *P. putida*.

Critically, the same study showed that "Both these organisms used direct sulfhydrylation of O-succinylhomoserine for the synthesis of methionine," meaning the dominant methionine route in *P. putida* is direct sulfhydrylation, not transsulfuration. PP_4348 therefore provides a *supplementary* transsulfuration route producing homocysteine (subsequently methylated to methionine by MetE/MetH), important particularly when cysteine or sulfur flux favors the cystathionine intermediate. Because *P. putida* lacks cystathionine γ-lyase (no reverse transsulfuration), the enzyme acts unidirectionally in the forward, cysteine → methionine biosynthetic direction. This regulatory and pathway context — low basal expression, cysteine inducibility, forward direction only — precisely defines PP_4348's metabolic niche.

Finding 3 — Catalytic mechanism, substrate positioning, and cytoplasmic localization

Studies of *E. coli* CBL (Clausen et al., 1996/1997) show a fold-type I PLP enzyme with three domains per monomer that assembles into a **homotetramer of active dimers**: "A homotetramer with 222 symmetry is built up by crystallographic and non-crystallographic symmetry. Each monomer of CBL can be described in terms of three spatially and functionally different domains" PMID: 8831789. Catalysis begins with the PLP forming an internal aldimine (Schiff base) with the active-site lysine (Lys186 in PP_4348); binding of substrate converts this to an external aldimine, and α,β-elimination cleaves the Cβ–S bond of cystathionine PMID: 9165088.

The determinant of *reaction-type selection* within the Cys/Met family is understood at the residue level. Messerschmidt et al. (2003) explain that "In CBLs this position is occupied by either tyrosine or hydrophobic residues directing binding of CTT [cystathionine] such that S is in the gamma-position" PMID: 12715888. In γ-family enzymes the equivalent position is a conserved glutamate; in β-lyases a tyrosine/hydrophobic residue reorients the substrate so that its sulfur sits in the γ-position, which geometrically dictates β-elimination rather than γ-chemistry. This is the molecular basis for why PP_4348 performs β-elimination on cystathionine.

Regarding localization, Q88EV4 has no signal peptide or transmembrane segment, and the transsulfuration pathway is cytosolic; PP_4348 is therefore a **soluble cytoplasmic protein** that carries out its function in the cytoplasm. The enzyme (and its bacterial homologs) also displays promiscuous cysteine-S-conjugate β-lyase activity (UniProt GO:0047804), a hallmark of the broad substrate tolerance of the family.

Finding 4 — All catalytic and active-site residues are conserved in PP_4348 (structure-based evidence)

A residue-by-residue alignment of PP_4348 (Q88EV4) to *E. coli* MetC (P06721) shows one-to-one conservation of every key active-site residue defined by the *E. coli* crystal structure (Clausen et al., 1996):

Function	*E. coli* MetC residue	PP_4348 residue
Catalytic PLP Schiff-base lysine	Lys210	**Lys186**
Substrate-orienting tyrosine	Tyr56	Tyr35
Active-site tyrosine	Tyr111	Tyr91
Binds PLP pyridine nitrogen	Asp185	Asp161
Active-site threonine	Thr222	Thr201
Substrate-carboxylate region tyrosine	Tyr338	Tyr312

The PLP-Schiff-base lysine mapping (Lys210 → Lys186) exactly matches the independently annotated UniProt PLP-lysine feature for Q88EV4, providing internal cross-validation. The *E. coli* structure defines the PLP-binding second domain: "The second domain (residues 61 to 256) harbors PLP and has an alpha/beta-structure with a seven-stranded beta-sheet as the central part" PMID: 8831789. The intact, colinear active-site constellation demonstrates that PP_4348 possesses a functional, canonical CBL catalytic machinery.

Finding 5 — KEGG pathway placement, EC-number nuance, and genomic context

KEGG (ppu:PP_4348) maps the gene to pathway **ppu00270** (Cysteine and methionine metabolism), **ppu00450** (Selenocompound metabolism), **ppu01230** (Biosynthesis of amino acids), and **module M00017** (Methionine biosynthesis, aspartate ⇒ homoserine ⇒ methionine), confirming its physiological placement at the cystathionine → homocysteine step. Its Pfam architecture includes Cys_Met_Meta_PP, Beta_elim_lyase, and Aminotran_5 motifs — all consistent with a fold-type I PLP enzyme.

A notable annotation nuance: KEGG assigns KO **K01760 / EC 4.4.1.13** (cysteine-S-conjugate β-lyase) whereas UniProt annotates **EC 4.4.1.8** (cystathionine β-lyase). Both are carbon-sulfur lyase (subclass 4.4.1) reactions that cleave a Cβ–S bond and are catalyzed by the same active site; the discrepancy reflects the enzyme's catalytic promiscuity, not a different enzyme.

The gene occupies KT2440 genome position ~4,940,549–4,941,643 and is **not embedded in a classical methionine operon**. It is flanked by PP_4347 (pseudogene), PP_4349 (conserved hypothetical), PP_4346 (D-Ala–D-Ala ligase), and PP_4350 (a NifS/IscS-family cysteine desulfurase). Its standalone genomic arrangement is consistent with the low basal, sulfur-source–regulated expression reported by Vermeij & Kertesz (1999).

Finding 6 — Substrate/reaction-specificity determinants are experimentally defined and conserved

Site-directed mutagenesis of *E. coli* CBL (Lodha & Aitken, 2011) experimentally established the functional roles of individual active-site residues in the α,β-elimination of L-cystathionine. The study confirmed the exact reaction: "Cystathionine β-lyase (CBL) catalyzes the hydrolysis of L-cystathionine (L-Cth) to produce L-homocysteine, pyruvate, and ammonia" PMID: 21958132. It further demonstrated "A novel role for residue S339 as a determinant of reaction specificity, via tethering of the catalytic base, K210" — with the S339-to-Ala variant reducing kcat/Km ~1000-fold. Tyr238 was shown to interact with the substrate's distal carboxylate and Tyr56 with the distal amino group.

In PP_4348, the corresponding catalytic base (Lys210 → Lys186) and the specificity tyrosines (Tyr56 → Tyr35, Tyr111 → Tyr91, Tyr338 → Tyr312) are all conserved. This places the experimentally validated *E. coli* specificity determinants directly onto the PP_4348 active site.

The family's promiscuity is well documented: bacterial MetCs display alliinase activity PMID: 38810088 and participate in D-amino acid metabolism PMID: 29592871. These secondary activities are consistent with the broad Cβ–S lyase capability of the family, but L-cystathionine β-elimination remains the physiological reaction of PP_4348.

Mechanistic Model / Interpretation

Position in methionine biosynthesis

PP_4348 catalyzes the penultimate step of methionine biosynthesis via the transsulfuration branch:

```
                 (sulfur assimilation)
                          │
                          ▼
    L-cysteine ──────────────────────────┐
                                          │  (cystathionine γ-synthase, MetB)
    O-succinyl-L-homoserine ─────────────►│
                                          ▼
                        L,L-CYSTATHIONINE
                                          │
                                          │  ★ PP_4348 = cystathionine β-lyase (MetC)
                                          │     EC 4.4.1.8, PLP-dependent
                                          │     Cβ–S bond cleavage (α,β-elimination)
                                          ▼
               L-HOMOCYSTEINE  +  pyruvate  +  NH₄⁺
                                          │
                                          │  (MetE / MetH methyltransferase)
                                          ▼
                             L-METHIONINE
                                          │
                                          ▼
                          S-adenosylmethionine
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In *P. putida*, this transsulfuration route runs in parallel with — and is subordinate to — the dominant **direct sulfhydrylation** of O-succinylhomoserine. Transsulfuration is upregulated when cysteine is the sole sulfur source, situating PP_4348 as a conditionally important, cysteine-inducible enzyme. Because *P. putida* has no cystathionine γ-lyase, the pathway is one-directional (biosynthetic; cysteine → methionine).

Catalytic mechanism

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Internal aldimine (resting)          External aldimine (substrate-bound)
 PLP=N—Lys186  ────────────────►  PLP=N—[cystathionine]  ────────────────►
              (transaldimination)             (Cα deprotonation →
                                               α,β-elimination at Cβ—S)
                                                      |
                                                      ▼
                        release of L-homocysteine (thiol leaving group)
                        + aminoacrylate—PLP intermediate
                                                      |
                                        (hydrolysis / reprotonation)
                                                      ▼
                   pyruvate + NH₄⁺ + regenerated PLP=N—Lys186
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The reaction specificity — β-elimination rather than the γ-elimination performed by cystathionine γ-synthase/γ-lyase — is dictated by active-site geometry. In β-lyases, a tyrosine/hydrophobic residue (in place of the conserved glutamate of γ-enzymes) orients cystathionine so that its sulfur occupies the γ-position, positioning the Cβ–S bond for cleavage. This is the structural switch, conserved in PP_4348, that defines its catalytic outcome.

Comparative summary

Property	Assignment for PP_4348	Basis
Primary reaction	L,L-cystathionine → L-homocysteine + pyruvate + NH_4^+	UniProt RHEA:13965; P 39644606 P 21958132
EC number	4.4.1.8 (UniProt); 4.4.1.13 also mapped by KEGG (promiscuity)	UniProt / KEGG K01760
Cofactor	PLP, Schiff base at Lys186	UniProt feature; alignment to Ec Lys210
Fold	Fold-type I PLP transferase, homotetramer	SSF53383; P 8831789
Pathway	Methionine biosynthesis (transsulfuration branch)	KEGG M00017; P 10482527
Direction	Forward (Cys → Met) only; no reverse transsulfuration	P 10482527
Regulation	Low basal; induced by cysteine as sole S source	P 10482527
Localization	Cytoplasm	No signal peptide in Q88EV4
Secondary activities	Cysteine-S-conjugate β-lyase, alliinase, D-AA metabolism	GO:0047804; P 38810088 P 29592871

Evidence Base

PMID	Title (abbrev.)	How it supports the annotation
10482527	*Pathways of assimilative sulfur metabolism in Pseudomonas putida*	**Organism-specific.** Directly measures cystathionine β-lyase in *P. putida*, shows cysteine-inducible transsulfuration and dominant direct sulfhydrylation. Anchors Finding 2.
8831789	*Crystal structure of E. coli cystathionine β-lyase at 1.83 Å*	Defines the fold, homotetramer, three-domain monomer, PLP-binding domain, and the catalytic residues mapped onto PP_4348. Anchors Findings 1, 3, 4.
9165088	*Mode of action of cystathionine β-lyase*	Establishes the PLP-dependent Cβ–S cleavage mechanism (internal/external aldimine, α,β-elimination). Supports Finding 3.
12715888	*Determinants of enzymatic specificity in the Cys-Met-metabolism PLP-dependent enzyme family*	Explains the tyrosine/hydrophobic vs. glutamate switch that makes CBLs perform β- rather than γ-elimination. Supports Finding 3.
21958132	*Side-chain hydroxyls Y56, Y111, Y238, Y338, S339 as specificity determinants in E. coli CBL*	Experimentally defines catalytic base K210 and specificity residues (all conserved in PP_4348) and confirms the exact reaction/products. Anchors Finding 6.
39644606	*Crystal structures of cystathionine β-lyase from Bacillus cereus*	Independent confirmation of the cystathionine → homocysteine + pyruvate reaction. Supports Finding 1.
38810088	*Identification and characterization of bacterial alliinase (MetC)*	Documents family promiscuity (bacterial MetCs with alliinase activity). Supports Finding 6.
29592871	*Cystathionine β-lyase is involved in D-amino acid metabolism*	Documents secondary D-amino-acid–related activity of CBL. Supports Finding 6.
10965031	*Crystal structure of L-methionine γ-lyase from P. putida*	Shows the related *P. putida* MGL shares the fold with CBL/CGS; context for family relationships and for distinguishing PP_4348 from *mdeA*.
17300162	*Inhibitors of bacterial cystathionine β-lyase*	Highlights that methionine biosynthesis is absent in mammals, making bacterial CBL an antibiotic

PMID	Title (abbrev.)	How it supports the annotation
		target — underscoring the enzyme's physiological importance.

Consistency of the evidence: Independent lines — UniProt/Rhea reaction annotation, diagnostic InterPro/Pfam signatures, quantitative pairwise alignment (43.9% identity to *E. coli* MetC, higher than any other family member), one-to-one active-site residue conservation including the PLP lysine, KEGG pathway/module placement, and direct enzymological measurement in *P. putida* itself — all converge on the same conclusion. No source contradicts the cystathionine β-lyase assignment.

Supported vs. Refuted Hypotheses

Supported - PP_4348 is a PLP-dependent cystathionine β-lyase (EC 4.4.1.8) — catalytic reaction, PLP-Lys186, IPR006233 signature, and sequence homology all agree. - It functions in the transsulfuration branch of methionine biosynthesis in the cytoplasm. - It is regulated (induced on cysteine) and operates in the forward direction only, given the absence of cystathionine γ-lyase in *P. putida*.

Refuted / excluded - PP_4348 is **not** the *P. putida* methionine γ-lyase (*mdeA*, P13254) — that is a separate gene; sequence identity to it (41.1%) is lower than to *E. coli* MetC (43.9%). - No evidence that PP_4348 acts in reverse transsulfuration (methionine → cysteine); *P. putida* lacks the requisite γ-lyase.

Limitations and Knowledge Gaps

1. **No direct biochemical characterization of the PP_4348 gene product specifically.** Vermeij & Kertesz (1999) measured cystathionine β-lyase *enzyme activity* in *P. putida* but did not link that activity to the PP_4348 locus by genetics or purification. The residue-level and reaction assignments for PP_4348 rest on high-confidence homology inference from *E. coli* MetC, not on a purified PP_4348 protein. Kinetic parameters (kcat, Km for L-cystathionine) for the PP_4348 enzyme are unknown.

2. **The Vermeij & Kertesz work used *P. putida* S-313, not strain KT2440.** These are closely related and the pathway logic is expected to be conserved, but strain-level regulatory differences are possible.
3. **EC-number ambiguity (4.4.1.8 vs 4.4.1.13).** UniProt and KEGG differ. The interpretation here — that this reflects catalytic promiscuity of a single active site — is well supported by family literature but has not been experimentally resolved for PP_4348 itself. The relative in vivo importance of the cysteine-S-conjugate β-lyase side activity is unquantified.
4. **Alignment identities among family members are close (41–44%).** The β-lyase assignment relies on the InterPro bacterial-CBL signature and the direction of the identity ranking rather than a large margin. A crystal structure or biochemical assay would remove any residual uncertainty.
5. **Regulatory circuitry not defined at the gene level.** The trans-acting regulators controlling PP_4348's cysteine-inducible, sulfur-source–dependent expression in KT2440 (e.g., a CysB/MetR-type regulator) have not been identified.
6. **Localization is inferred**, not experimentally fractionated — it rests on absence of secretion/membrane signals plus the cytosolic nature of the pathway.

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzymology.** Clone PP_4348 from KT2440, express with an affinity tag, purify, and measure steady-state kinetics for L,L-cystathionine (expected products: L-homocysteine, pyruvate, NH_4^+). Determine kcat and Km and compare to *E. coli* MetC. Assay UV-vis for the characteristic PLP internal aldimine (~420 nm).
2. **Substrate-specificity panel.** Test the purified enzyme against L-cystathionine, cysteine-S-conjugates, alliin/S-alk(en)yl-L-cysteine sulfoxides, djenkolate, and O-acetyl/O-succinylhomoserine to quantify the physiological reaction versus promiscuous activities, directly resolving the EC 4.4.1.8 vs 4.4.1.13 question.
3. **Genetic knockout and complementation.** Construct a ΔPP_4348 mutant in KT2440; test methionine auxotrophy/growth phenotypes under different sulfur sources (sulfate, cysteine, methionine). Predicted outcome: a conditional methionine requirement most pronounced when cysteine is the sole sulfur source, given the dominant direct-sulfhydrylation route.

4. **Active-site validation by mutagenesis.** Substitute Lys186 (catalytic PLP lysine) and the specificity tyrosines (Tyr35, Tyr91, Tyr312) to confirm the homology-based residue assignments and quantify their catalytic contributions, mirroring the *E. coli* Lodha & Aitken (2011) study.
5. **Structural determination or modeling.** Solve a crystal structure (or generate a validated AlphaFold model with PLP docked) to confirm the fold-type I homotetramer and the geometry that positions cystathionine sulfur in the γ-position for β-elimination.
6. **Expression profiling in KT2440.** Use RT-qPCR or RNA-seq across sulfur sources (sulfate vs. cysteine vs. methionine) to confirm cysteine-inducible expression of PP_4348 specifically in KT2440 and identify the responsible regulator.

Conclusion

PP_4348 (Q88EV4) of *Pseudomonas putida* KT2440 is a cytoplasmic, PLP-dependent **cystathionine β-lyase (MetC, EC 4.4.1.8)** that cleaves the Cβ–S bond of L,L-cystathionine to produce L-homocysteine, pyruvate, and ammonia — the transsulfuration step of methionine biosynthesis (KEGG M00017). The assignment is supported by convergent, mutually consistent evidence: diagnostic bacterial-CBL InterPro/Pfam signatures, full conservation of the experimentally validated *E. coli* MetC active site (including the PLP Schiff-base Lys186), correct pathway placement, and direct enzymological detection of cysteine-inducible cystathionine β-lyase activity in *P. putida*. In this organism the enzyme runs only in the forward, cysteine → methionine direction (no reverse transsulfuration) and supplements the dominant direct-sulfhydrylation route. The gene symbol, organism, and protein family are unambiguous — there is no confounding gene-symbol conflict.
